

Degrees of separation: graduate earnings track institutional status more than the academy’s departmental hierarchy

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Abstract

The academy ranks its own departments through faculty hiring. We ask whether this hierarchy carries information about bachelor’s graduates’ earnings, field by field. Using public data only, we measure within-field coupling, the rank correlation across institutions between a department’s position in the US faculty-hiring hierarchy and its graduates’ median earnings, for 52 US fields (College Scorecard, and Census PSEO as an overlapping second source) and, with a UK hiring hierarchy built from ORCID-derived records, for 33 UK subjects (DfE LEO). The analysis is descriptive. Coupling is positive and varies across fields, but most of it is shared with institution-level status: controls for selectivity, Pell share, control and state lower mean US coupling from +0.43 to +0.14. Within institutions, net of each field’s earnings slopes on selectivity and academia-wide prestige, one standard deviation of department prestige goes with 0.8% higher four-year pay, a remainder whose 95% interval reaches zero. It is individually clear only in computer science, on a sample without the University of California campuses and several flagship public departments. Field prestige predicts pay no better than academia-wide prestige. In the UK most of the prestige ordering of pay survives a direct control for graduates’ prior attainment. Within fixed graduation cohorts coupling rises by about 0.03 per year in both countries. In the US it also rises across cohorts, so its career-time part can be bounded only under sign assumptions; about three fifths of the within-cohort rise is shared with selectivity; and net of selectivity measured on recent entrants, a dating that favours this result, the loading of pay on academia-wide status does not decline within cohorts. Coupling is weak in UK subjects paid on national scales. The academy’s departmental hierarchy adds little labour-market information beyond institution-wide status.

Keywords: faculty hiring networks; academic prestige; graduate earnings; college selectivity; field of study; administrative data; career dynamics; science of science

1 Introduction

The academy ranks its own departments. Each time a department hires a professor it reveals which doctoral programmes it trusts to train faculty, and aggregated over a decade these choices form a steep hierarchy in which a few institutions train most of the faculty in their field (Clauset et al., 2015; Wapman et al., 2022b). The ordering comes from peers’ hiring decisions, not from admissions statistics, reputation surveys or composite rankings, and it exists field by field. We ask whether it carries information about how the academy’s bachelor’s graduates fare outside it: within a field, do graduates of departments higher in the hiring hierarchy earn more, and does the association belong to the department or to the institution around it?

The question matters for two literatures. For the science of science, the hiring hierarchy is a carefully measured status ordering, and position in it shapes productivity, the spread of ideas and who becomes faculty (Way et al., 2019; Morgan et al., 2018, 2022; Zhang et al., 2022); whether it is also legible outside academia, where nearly all bachelor’s graduates work, has not, to our knowledge,

been measured. For the economics and sociology of higher education, it sharpens an old question. Earnings rise with institutional selectivity and brand, and much of that gradient is attributed to who enrolls (Dale and Krueger, 2002, 2014; Mountjoy and Hickman, 2021); the gradient differs by field of study (Eide et al., 2016; Quadlin et al., 2021; Hastings et al., 2013; Bloem et al., 2024); and in the United Kingdom an institution’s academic standing adds little information about graduates’ returns beyond the selectivity of its intake (Britton et al., 2022). A department-level, peer-defined prestige axis asks how much of the status gradient in pay belongs to the department, and how the gradient changes over careers, where employer-learning models (Farber and Gibbons, 1996) predict that the weight on an easily observed credential falls once a hard-to-observe correlate of productivity is held fixed (Altonji and Pierret, 2001).

We measure within-field coupling: the Spearman rank correlation, across the institutions that teach a field, between a department’s position in the US faculty-hiring hierarchy (Wapman et al., 2022b) and the median earnings of its bachelor’s graduates. We map it across 52 US fields on College Scorecard earnings (U.S. Department of Education, 2026), repeat it on Census Post-Secondary Employment Outcomes (PSEO) (U.S. Census Bureau, Longitudinal Employer-Household Dynamics, 2025, 2026), and, for the United Kingdom, build an academia-wide hiring hierarchy from ORCID-derived records of PhD-to-faculty moves (Li, 2026) and set it against Longitudinal Education Outcomes (LEO) earnings, which are also released by graduates’ prior attainment (Department for Education, 2026). All inputs are public, and every result is descriptive.

Four findings hold. First, coupling is positive, varies across fields and has a similar field ordering on the two US earnings sources. Second, most of it is shared with institution-level status: controls for selectivity, Pell share, control and state lower mean US coupling from +0.43 to +0.14; within institutions one within-field standard deviation (SD) of department prestige goes with 0.8% higher pay, a remainder whose interval reaches zero; and a field’s own prestige ordering predicts pay no better than the academia-wide ordering. Third, in the UK most of the coupling survives a direct control for graduates’ prior attainment; among graduates of the same attainment band, what survives tracks intake selectivity at least as closely as hiring-network prestige if the UK hierarchy is measured with little error. Fourth, within fixed graduation cohorts coupling rises with years since graduation in both countries; in the US it also rises across cohorts, so its career-time part can only be bounded, and much of the within-cohort rise is shared with selectivity. Coupling is weak in UK subjects on national pay scales and falls across US fields with the share of graduates in strictly licensed occupations, contrasts between discipline families that public data cannot separate from field type or employer sector.

Much of this is known in kind (Section 2). What is new is (i) the faculty-hiring hierarchy as the prestige axis, which separates a department’s standing in its field (field prestige, F) from its institution’s standing in the academy as a whole (academia-wide prestige, G), and with it a within-institution test of whether higher-standing departments inside the same institution have better-paid graduates; (ii) fixed-cohort estimates, in two countries, of how coupling changes with years since graduation, new in design and scope rather than in direction (Thomas and Zhang, 2005; MacLeod et al., 2017), which in the US load on academia-wide rather than field prestige in a weakly identified decomposition; and (iii) measurement: a bracket on the sampling reliability of the published ranks, a check against noise in the program medians, and a UK academia-wide hiring hierarchy. The US field map is a replication in kind: its ordering is close to the known ordering of fields by how steeply pay follows selectivity (Bloem et al., 2024), and similar on a second, overlapping earnings source.

For the academy’s departmental hierarchy the answer is mostly negative: it tracks graduate pay largely through the institutional status it shares with selectivity, and beyond that goes with a small remainder, borderline on average and individually clear only in computer science, which public data cannot attribute to department standing or to program-level selection. The computer-

science estimate describes a sample that lacks the University of California campuses and Caltech, which have no recent SAT average, and several flagship public departments lost in the name join, Georgia Tech, Purdue, Ohio State and the University of Washington among them (Section 4.2). Further methods and all robustness detail are in the Supplementary Information (SI).

2 Background and positioning

College quality and selection. Once application behaviour absorbs selection, the average earnings payoff to a more selective college is small, with larger returns for disadvantaged students (Dale and Krueger, 2002, 2014); other designs find sizeable effects in some settings (Hoekstra, 2009; Brewer et al., 1999; Witteveen and Attewell, 2017). Single quality proxies are noisy (Black and Smith, 2006), matching estimates depend on common support (Black and Smith, 2004), returns differ across the earnings distribution (Andrews et al., 2016), raw differences shrink once selection is controlled (Cunha and Miller, 2014), and the payoff to elite colleges concentrates in the upper tail (Zimmerman, 2019; Chetty et al., 2026). Most of this work measures standing by selectivity, rankings, brand or value added (Riehl et al., 2019; Mountjoy and Hickman, 2021). The closest academic measure is Hussain et al. (2009): in UK graduate surveys an institution’s research rating predicts wages but correlates 0.87 with entry tariff and turns negative, imprecisely, alongside other quality measures.

Field of study and status by field. Returns to field are large and heterogeneous (Kirkebøen et al., 2016; Altonji et al., 2012, 2016a; Webber, 2014; Kim et al., 2015; Andrews et al., 2024), and so is the status gradient within field (Eide et al., 2016; Quadlin et al., 2021; Hastings et al., 2013). Closest to us, Bloem et al. (2024) show on an earlier Scorecard release that median pay barely varies with admit rate in nursing, teacher education, social work and criminal justice but varies strongly in computer science, finance, economics, political science and marketing; cross-classified models find selectivity matters more outside STEM (Muse and Muse, 2024). UK returns by institution and subject were estimated on surveys (Walker and Zhu, 2011, 2018) and now on administrative LEO records (Belfield et al., 2018; Britton et al., 2019, 2020). The closest design to ours is Britton et al. (2022): within subject, returns at age 30 track the prior attainment of a degree’s intake closely in law, business and economics and not at all in medicine or education, which the authors attribute to centrally regulated wages, and net of selectivity no subject’s league-table rating correlates positively and significantly with returns.

Faculty-hiring hierarchies. Centrality in PhD-exchange networks tracks departments’ reputational prestige (Burris, 2004); hiring networks are steeply hierarchical (Clauset et al., 2015; Wapman et al., 2022b); PhD origin is tightly linked to placement (Fowler et al., 2007; Headworth and Freese, 2016); placement records have been used to rank departments (Amir and Knauff, 2008); and networks divide into clusters with little hiring between them (Terviö, 2011). The hierarchy is estimated with SpringRank, which orders institutions so that most hires flow down the ordering (De Bacco et al., 2018). This literature measures standing within academia; to our knowledge it has not asked whether position in the hierarchy predicts the pay of the academy’s bachelor’s graduates, a question made less than obvious by the weak link between research and teaching quality (Hattie and Marsh, 1996).

Employer learning and career dynamics. In employer-learning models, the wage coefficient on a credential that employers see at hiring need not change with experience when a hard-to-observe

correlate of productivity is left out (Farber and Gibbons, 1996); with that correlate in the model, the credential’s coefficient falls with experience while the correlate’s rises (Altonji and Pierret, 2001). Employers learn quickly on average (Lange, 2007), but the speed varies by occupation (Mansour, 2012), and learning about a productivity that keeps changing stays imperfect over whole careers (Kahn and Lange, 2014); for college graduates the loading on test scores is already high at entry and grows little (Arcidiacono et al., 2010). Among students who applied to and were admitted by similar colleges, the premium to a more selective one is small over the career (Dale and Krueger, 2014) or fades within a few years (Mountjoy and Hickman, 2021); prestige premia fade with experience in Chile (Bordón and Braga, 2020) and matter less for later promotions in Japan (Araki et al., 2016), while US graduates of higher-quality colleges have faster early earnings growth (Thomas and Zhang, 2005) and college reputation correlates with earnings growth in Colombia (MacLeod et al., 2017). Entry conditions and field-specific forces shape careers too (Oreopoulos et al., 2012; Roksa and Levey, 2010; Altonji et al., 2016b; Deming and Noray, 2020); separating career time from calendar time and cohort requires assumptions (Fosse and Winship, 2019a,b), and administrative earnings records lose graduates selectively (Foote and Stange, 2026).

What is known and what is new. Known: small average returns to selectivity net of selection, elite effects in the tails, fast average employer learning, a college-quality earnings gradient that grows early in careers, and status gradients that vary by field, with little gradient where wages are set centrally (Britton et al., 2022; Bloem et al., 2024). Our field map and UK check, which uses a coarser, band-level attainment control than Britton et al. (2022), are replications in kind (Section 1). Licensing and pay schedules enter only in Section 4.5, as candidate readings of where coupling is weak.

3 Data and measurement

Table 1 lists each source with its unit, the sample used and its known limitations; versions, URLs and access dates are recorded in the repository (SI Section S1).

3.1 Prestige: the faculty-hiring hierarchy

We use the published prestige ranks of Wapman et al. (2022b), estimated with SpringRank from a census of faculty at 368 PhD-granting US institutions, 2011–2020 (Wapman et al., 2022a). For institution i and field f , field (department) prestige F_{if} is minus the published field rank, and academia-wide prestige G_i is minus the rank in the network that pools hires across all fields. Both are fixed across earnings horizons and describe hires of 2011–2020. G tracks selectivity: across the 242 universities of the Opportunity Insights sample its Spearman correlation with average SAT is +0.68, and the UK analogue correlates +0.71 with intake selectivity across 133 providers. Within the field-by-cohort cells of the PSEO panel F and G correlate 0.83 on average. F partly tracks doctoral output: residualising it on log research doctorates and the net export ratio of hires keeps 0.60 [0.47, 0.73] of mean coupling, an upper bound on the production share (post hoc, self-attested registration; SI Section S2).

The public release omits institution pairs linked by a single hire, so a SpringRank rebuilt from public edges reproduces the published field ranks only partly (median Spearman 0.73; academia-wide 0.908). We therefore use the published ranks and bracket their sampling reliability: the split-half reliability of F on public edges averages 0.81 across 57 fields (stepped up with the Spearman–Brown formula; Spearman, 1910; Brown, 1910), a lower bound, and 0.96 when extrapolated to the

number of hires behind the published ranks; that of G is 0.99 (SI Section S2). Where reliability matters, results are reported at both ends.

3.2 Earnings and placement

US: College Scorecard. The canonical outcome is the median earnings of Title IV–aided bachelor’s completers four years after completion, from the Field-of-Study file released in June 2026 (U.S. Department of Education, 2026), which excludes completers enrolled or not working in the measurement year, aggregated over each field’s four-digit CIP codes through a curated map of 66 fields. Matching to the published ranks by normalised institution name gives 3,444 institution-by-field cells in the 52 fields with at least 15 matched institutions (median 48, range 15–173), the canonical sample, and 3,500 cells in the 57 fields with at least 8, used for heterogeneity, reliability and cross-source statistics. Scorecard covers 66–74% of the ranked institutions in each academia-wide prestige quartile (SI check, SI Section S1), but the name join loses institutions whose Scorecard names carry campus qualifiers (Georgia Tech, Purdue and Ohio State among them). Institutional covariates (average SAT, admission rate, Pell share, control, institution-wide earnings and the state earnings level) come from the “most recent” institution file, which describes recent entering classes; its SAT average is missing for test-blind institutions (the University of California campuses and Caltech among them), and for three key statistics we re-date the controls to the graduates’ entry years with the historical files. No public US file measures the entry attainment of a program’s own students. One- and five-year medians refer to other cohorts; they serve as replications and enter averages over the three horizons.

US: PSEO. PSEO reports unemployment-insurance wage earnings of graduates with earnings above a full-time minimum-wage threshold, in constant 2023 dollars, one, five and ten years after graduation (y1, y5, y10) by three-year graduation cohort. Release V4.13.0 (2025Q4) is canonical; the August 2026 release V4.14.1 (2026Q2), on which the later analyses run, is a re-check. For career time we keep, within each field and cohort (graduates of 2001–03 to 2010–12), the institutions observed at all three horizons, at least 15 per cell: 106 institutions, 142 field-by-cohort cells and 39 fields on V4.13.0 (116, 145 and 40 on V4.14.1). A window-by-horizon surface adds later cohorts: 17 cells, from 2001–03 to 2019–21 at y1, to 2016–18 at y5 and to 2010–12 at y10. PSEO partners are mostly public: only 20 of the 93 top-quartile ranked institutions have a usable five-year cell on V4.13.0 (26 on V4.14.1). PSEO Flows give each institution-by-CIP-2 cell’s employed graduates by employer sector (SI Section S7).

UK: LEO and an academia-wide hiring hierarchy. The DfE provider-level LEO file reports median earnings by provider, CAH2 subject, graduating cohort and year after graduation (YAG 1, 3, 5), separately by prior-attainment band, sex and POLAR4 quintile (one characteristic at a time) (Department for Education, 2026). We use the cohorts 2013/14 to 2016/17 and the 33 subjects with at least 15 providers. For prestige we run SpringRank on 11,987 PhD-to-faculty moves between institutions in Great Britain in the ORCID-derived academic mobility edges (Li, 2026; ORCID, Inc., 2026), matched to providers through the ROR data dump (Research Organization Registry, 2026): 134 providers form the primary sample (SI Section S1). Russell Group members sit at a mean percentile of 0.86 against 0.43 for other providers, and the split-half reliability is 0.92. Validity is less certain: in the US, an academia-wide ranking rebuilt from the ORCID-derived edges agreed with the published ranks at Spearman 0.74 (SI Section S2). Field-tagged UK networks are too sparse to use, so the UK analysis has no department axis. Intake selectivity is the institution’s share of graduates who entered with at least 360 tariff points.

Other sources. Opportunity Insights college tables (Chetty et al., 2020) give tail outcomes for 242 PhD-granting universities. The 2023 American Community Survey public-use microdata (U.S. Census Bureau, 2026) give field-level licensed and sector shares, graduate-degree shares and field-by-state earnings. The Survey of Earned Doctorates (National Center for Science and Engineering Statistics, 2026) gives research doctorates by baccalaureate institution, state unemployment rates come from the Bureau of Labor Statistics (U.S. Bureau of Labor Statistics, 2026), and admission rules for computer-science majors were coded from official university pages (Table 1).

3.3 Coupling and inference

Coupling is $\rho_f = \text{Spearman}_i(F_{if}, Y_{if})$ across the institutions i that teach field f , with Y median earnings: a rank statistic that asks whether the prestige and pay orderings coincide, not how large the pay differences are. Partial couplings correlate the rank residuals of F and Y after regressing both on the ranks of the controls. The main text rests on three estimands: coupling and partial coupling averaged over fields; the within-institution slope β of log earnings on department prestige (Eq. 1), a pay difference; and the per-year change of coupling within fixed graduation cohorts. A field is flagged reliable when its earnings carry cross-institution signal beyond sampling noise, its interval excludes the band that noise alone would produce under perfect agreement, and it has at least 10 institutions; 20 of the 52 canonical fields pass, and the flag is a check, not a filter (SI Section S3).

Fields share institutions, so cross-field statistics are not independent. Means across fields on Scorecard use a joint institution bootstrap, conditional on the observed fields; PSEO statistics use the two-way (field, institution) cluster variance (Cameron et al., 2011), with the conservative crossed field-by-institution bootstrap beside it; within-institution slopes use an institution cluster bootstrap; UK intervals resample subjects and then providers; tests clustered by state use the wild cluster bootstrap (Cameron et al., 2008; Webb, 2023); field-level moderators are tested by field-label permutation and by resampling whole CIP-2 families. Brackets are 95% intervals. Where interval types disagree, an association counts as present only if its interval excludes zero under the most conservative variance reported for it. Four statements fail that rule and are reported as not established (SI Section S9 lists their intervals): the within-institution slope, with 2026 or with entry-year selectivity controls; the gap between the growth of the G and F loadings; and the UK per-cell partial of G net of selectivity on all-graduate medians. The canonical specification was fixed after the main analyses had been run and before drafting; alternatives form a specification curve (Simonsohn et al., 2020) (SI Section S4). The analyses added later (scripts 65–69 and 72–75) wrote their tests and reading rules into the script before estimation, a self-reported timing; the SI lists the amendments made after a first run, and SI Section S9 every hypothesis examined in scripts 52–75, including the verdicts of all nine registered tests of script 75, of which only two enter the text, with the power or interval of the main nulls. Every number comes from a seeded script or, where marked “SI check”, from code run on the repository’s tables.

4 Results

We report the four findings in turn and then the fields where coupling is weak. Table 2 collects the controls for institution-level status and Table 3 the career-time estimates.

Table 1: Data sources and samples. All sources are public; provenance is recorded in `data/raw/SOURCES.md` of the repository.

Source	Unit and release	Sample used here	Role	Known limitation
Faculty-hiring ranks (Wapman et al., 2022a)	institution $\times$ field; published SpringRank ranks, 2011–2020	66 fields; 52 with ≥ 15 institutions	F, G	single-hire pairs not public; reliability of F bracketed
College Scorecard, Field of Study (U.S. Department of Education, 2026)	institution $\times$ CIP-4; bachelor’s median earnings at 4 years (1, 5 as replications); June 2026	3,444 cells, 52 fields	Y (canonical)	Title IV completers; medians; no program-level entry attainment
College Scorecard, institution files (U.S. Department of Education, 2026)	“most recent” (2026) and historical files (2001–02 to 2016–17)	46–47 fields with SAT data	selectivity, Pell, control, state	institution-level; entry year inferred
PSEO earnings (U.S. Census Bureau, Longitudinal Employer-Household Dynamics, 2025, 2026)	institution $\times$ CIP-4 $\times$ 3-year cohort; p25, p50, p75 at y1, y5, y10; V4.13.0, V4.14.1	fixed cohorts 2001–2012; 17-cell surface, 24 fields	second source; career time	public-skewed; 20 (26) of 93 top-quartile institutions
PSEO Flows (U.S. Census Bureau, Longitudinal Employer-Household Dynamics, 2025, 2026)	institution $\times$ CIP-2 $\times$ employer sector	1,312 cells, 108 institutions	placement	noisy counts; industry, not occupation
DfE LEO provider file (Department for Education, 2026)	provider $\times$ CAH2 $\times$ cohort $\times$ YAG; by prior attainment, sex, POLAR4	33 subjects, 134 providers	UK earnings	marginal breakdowns only; coarse bands
ORCID-derived mobility edges (Li, 2026); ROR (Research Organization Registry, 2026)	PhD-to-faculty moves within Great Britain	11,987 moves, 551 institutions	UK G	self-reported careers; academia-wide only
Opportunity Insights (Chetty et al., 2020)	college; outcomes by parent income; Barron’s 2009 index	242 universities	tail outcomes	pools a college’s fields
ACS PUMS 2023 (U.S. Census Bureau, 2026)	person	field and field $\times$ state aggregates	shares; local demand	state of residence
Survey of Earned Doctorates (National Center for Science and Engineering Statistics, 2026); IPEDS (National Center for Education Statistics, 2026)	baccalaureate institution	PSEO panel institutions	doctorate rate	institution-wide
BLS LAUS (U.S. Bureau of Labor Statistics, 2026); admissions pages	state $\times$ year; CS program	18 states; 88 of 128 programs	entry conditions; entry route	residence state; 2026 policy
Occupational prestige (Hughes et al., 2024); NY Fed data by major (Federal Reserve Bank of New York, 2026)	field	32–52 fields	placement check	field level only

4.1 Coupling is positive, heterogeneous and similar on two earnings sources

Graduate pay tracks the hiring hierarchy in most fields. Across the 52 canonical fields mean coupling is $+0.419$ [$+0.355$, $+0.482$], and 39 fields have coupling more than 1.96 bootstrap standard errors above zero, none below (Fig. 1). Coupling varies across fields by more than institution sampling produces: in a three-level meta-analytic model of Fisher- z coupling on 57 fields (Konstantopoulos, 2011; Cheung, 2014), Cochran’s $Q = 183.2$ on 56 degrees of freedom and $I^2 = 0.69$ (Higgins and Thompson, 2002). Part of this spread is noise in the program medians; a secondary analysis, not yet independently verified, that propagates it lowers I^2 to 0.51 and keeps the field ordering (SI Section S3). Part of the remaining variation is a measurement confound: coupling rises with the number of institutions that teach a field (Spearman 0.43, 57 fields), and a discipline (CIP-2) variance component, detected without adjustment, is borderline once field size is a covariate (permutation p from 0.028 to 0.046; SI Section S3). The grouping of fields into disciplines is therefore a descriptive map.

The field ordering is similar on a second earnings source: coupling from PSEO five-year earnings correlates with Scorecard coupling across fields at $+0.68$ (two-way interval [$+0.54$, $+0.82$]; 57 fields, V4.13.0) and $+0.74$ [$+0.61$, $+0.87$] on V4.14.1. This is robustness to the earnings measure, not an independent replication: both use the same F , and 61% of the PSEO rows used on V4.14.1 also enter the Scorecard couplings. Pay also tracks where graduates go: within CIP-2 families in PSEO Flows, family prestige orders the share of employed graduates in market sectors about as closely as it orders pay ($+0.40$ [$+0.32$, $+0.47$] against $+0.41$ [$+0.31$, $+0.49$]; 1,312 cells; SI Section S10).

4.2 Most of the coupling is shared with institution-level status

Controlling for institution-level status removes most of the coupling, and what remains is small and borderline on average. The estimand is the information in F about pay net of institution-level status proxies, not a return. Average SAT is a candidate quality treatment as well as a selection proxy (Dale and Krueger, 2002), so partialling it out removes selection together with any college effect correlated with it: the drop measures what cannot be separated from institutional status with public data, not what selectivity causes.

The control ladder. On a common sample of 44 fields, mean coupling falls from $+0.43$ raw to $+0.31$ with Pell share, control and the state earnings level, to $+0.24$ with the admission rate and to $+0.14$ with average SAT (Fig. 2a; Table 2). On the 46 fields estimable with the broad set (SAT, admission rate, Pell share, control and state earnings level) the mean partial coupling is $+0.141$ [$+0.075$, $+0.208$], and 11 fields stay more than 1.96 standard errors above zero. Dating the controls to the falls in which the earnings cohort entered (2014–15) leaves the broad partial where it was (difference $+0.005$ [-0.031 , $+0.041$] on the same 2,787 programs, two-way; SI Section S4). Adding G to the broad set leaves $+0.059$ [$+0.003$, $+0.114$] (44 fields), the part of the department ordering that neither selectivity nor the academia-wide ordering tracks; across ten specifications that net out selectivity and G , its interval excludes zero in four.

Within institutions. Institution fixed effects absorb each institution’s average earnings level across its fields, and with it the institution-wide parts of selectivity, G and location. The slope of log four-year earnings on department prestige falls from $+0.084$ (SE 0.006) to $+0.013$ per within-field SD when they are added (3,416 programs, 198 institutions; Fig. 2b). With field-specific slopes on SAT, admission rate, Pell share and G (Eq. 1) it is $+0.0084$ [$+0.0000$, $+0.0164$] (2,889 programs, 168 institutions, 47 fields; institution cluster bootstrap; [-0.0012 , $+0.0180$] under the two-way

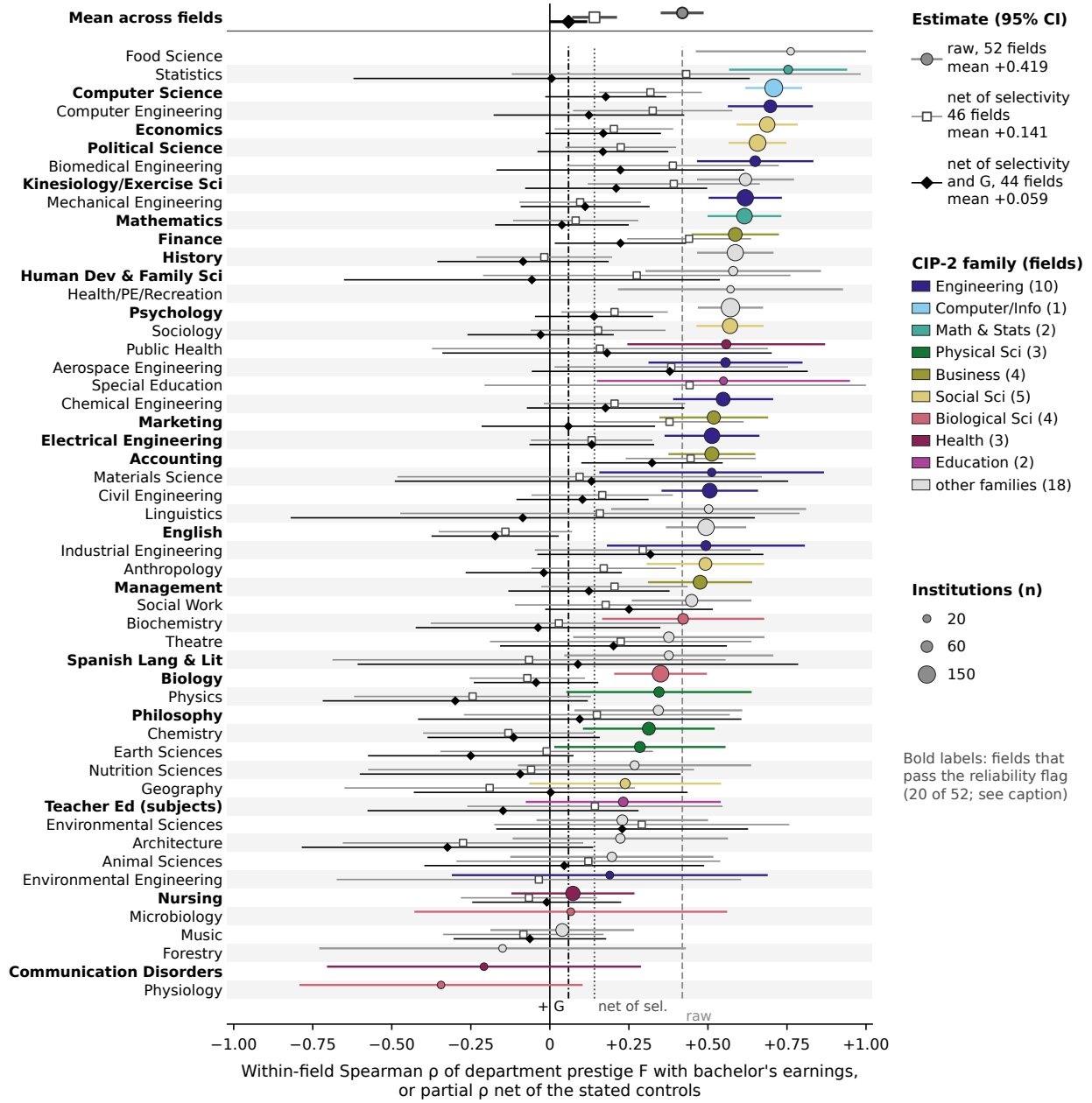


Figure 1: Coupling by US field: the Spearman correlation ρ_f across institutions between published field prestige F (Wapman et al., 2022b) and Scorecard median earnings four years after completion, for the 52 fields with at least 15 matched institutions. Circles: raw ρ_f , coloured by CIP-2 family, sized by institutions. Squares: net of the broad selectivity set (46 fields); diamonds: net of G as well (44 fields). Intervals: ± 1.96 joint institution-bootstrap SE, cut at ± 1 ; top row: means with 95% intervals. Bold labels: the 20 fields flagged reliable (Section 3.3); the flag does not certify precision.

variance): one SD of department prestige goes with 0.84% higher four-year pay, with an upper 95% limit of 1.6–1.8% and a lower limit at or below zero. With one- and five-year medians, which describe other cohorts, the slope is about zero (-0.0024 and $+0.0023$); averaged over the three horizons it is $+0.0048$ [-0.0024 , $+0.0120$] with programs weighted equally and $+0.0108$ [$+0.0035$, $+0.0181$] weighted by graduates. The point estimate barely moves with selectivity dated to entry years ($+0.0081$ against $+0.0090$ on the same 2,801 programs), with cubic field-specific slopes in G ($+0.0093$) or with a field-by-state earnings control for local demand (change $+0.0000$ [-0.0007 , $+0.0008$]), but that control is weak: it describes the institution’s state and passes little through to program earnings (SI Section S4). Controlling for program composition does not lower the slope either: on the 1,294 programs that release a Pell share (141 institutions, 26 fields), a field-specific control for it moves the slope from $+0.0118$ to $+0.0137$ [$+0.0027$, $+0.0238$] (change $+0.0019$ [-0.0022 , $+0.0053$]). A hint of composition within programs comes from one of several comparisons: with that control the slope is $+0.0009$ [-0.0123 , $+0.0128$] for aided graduates without a Pell grant (difference from all aided graduates $+0.0129$ [$+0.0008$, $+0.0243$]), but $+0.0102$ [-0.0024 , $+0.0222$] for Pell recipients (difference $+0.0035$ [-0.0079 , $+0.0152$]), and in the partial-correlation design the two subgroups look alike ($+0.049$ and $+0.048$, 24 fields; SI Section S4).

Department prestige against its competitors. In per-field rank regressions on 47 fields, the mean Shapley share of R^2 (Budescu, 1993; Grömping, 2007; Shorrocks, 2013) is 0.076 for F , against 0.173 for selectivity, 0.086 for G and 0.201 for institution-wide earnings, which contain the field’s own graduates (Fig. 2c). Before any control, G couples with field earnings slightly more strongly than F (mean $c_F - c_G = -0.049$ [-0.085 , -0.011], 57 fields), but F is measured with more error. Corrected for reliability (Spearman, 1904), the difference is -0.005 [-0.043 , $+0.044$] at the lower-bound and -0.042 [-0.077 , -0.001] at the extrapolated reliability, and net of selectivity it is $+0.00$ [-0.05 , $+0.05$] (Fig. 2d). Field prestige thus predicts pay no better than academia-wide prestige under every correction; whether it predicts pay worse depends on its reliability.

Where the remainder sits. Only computer science, one of three fields named in advance as high-coupling, has an individually clear remainder: $+0.050$ [$+0.012$, $+0.096$] per SD pooled over horizons (128 programs; $+0.007$, $+0.114$] correcting for the three fields). Whether per-field slopes differ at all depends on whether each field’s earnings gradient in G may be curved, and the concentration in computer science and engineering was read off the results (SI Section S4). The computer-science sample lacks 21 of the 50 highest-ranked departments: the six University of California campuses and Caltech, which have no fall-2024 SAT average; 11 lost in the name join, among them Georgia Tech, Purdue, Ohio State and the University of Washington; and 3 without a one- or five-year median. Entry-year SAT averages exist for the University of California campuses and Caltech, and a join by institution identifier would restore most of the others, but neither re-estimate was run, so the computer-science result describes this sample only. A test of whether the slope differs between programs that restrict entry to the major and programs that students can declare is too weak to separate selection into the major from department standing (80% power only against a difference of 0.187; SI Section S4).

Two readings. The residual prestige measure is noisy, and partialling G removes some genuinely field-specific variation, so the remainder is biased toward zero: disattenuated with the reliability bracket, the within-institution slope is $+0.0098$ to $+0.0297$. A coefficient-stability bound (Oster, 2019) puts the selection on unobservables needed to erase it at 0.20–0.67 times the selection on the observed controls ($R_{\max} = 1$), below the usual benchmark of 1 (SI Section S4). Under one

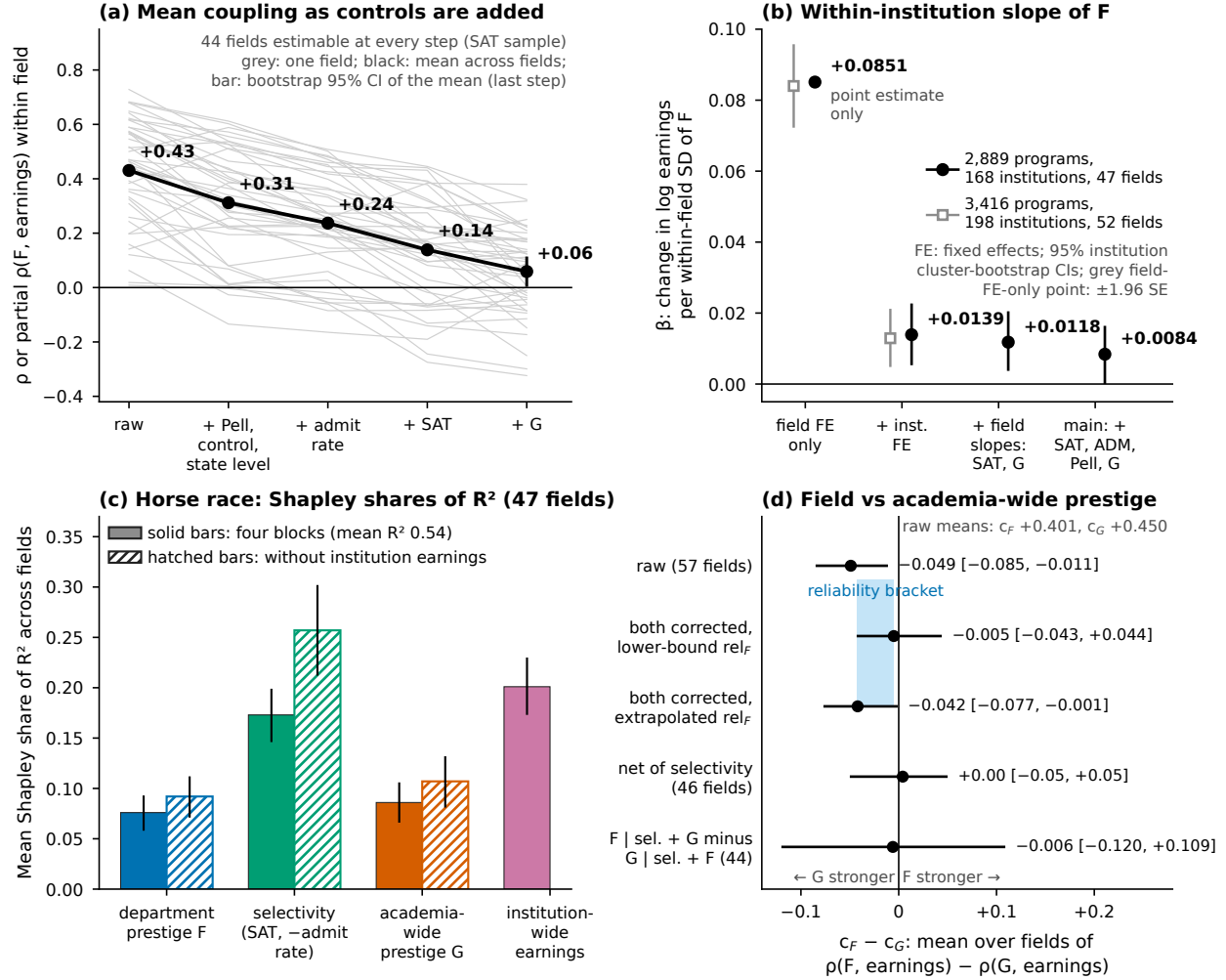


Figure 2: What remains of US coupling after institution-level status (Scorecard, four-year earnings). (a) Coupling of F as controls are added: fields (grey), mean (black; 44 fields). (b) Within-institution slope per within-field SD of F , from field fixed effects to the main specification (Eq. 1); black: 2,889 programs; grey: all 3,416. (c) Mean Shapley shares of R^2 in per-field rank regressions (47 fields), with and without (hatched) institution-wide earnings. (d) Mean $c_F - c_G$: uncorrected, corrected for error in F at either reliability, and net of selectivity. 95% institution-bootstrap intervals.

reading, the academy's field-specific hierarchy carries labour-market information of its own; under the other, coupling is the field's gradient in institutional status plus student quality that institution-level measures miss, for example selection into majors that admit on grades (Bleemer and Mehta, 2022). The bulk of the coupling fits the second; for the remainder, including the Pell/non-Pell contrast, the two predict the same pattern. The cross-field ordering does not separate them either: partial couplings still rank fields like raw couplings beyond shared estimation noise, but raw coupling and how steeply each field's earnings follow SAT have a true-score correlation of +0.94 (23 fields; SI Section S4), so both readings predict the same ordering. Nor does the upper tail: across 242 PhD-granting universities, net of selectivity and parental income, G 's association with the share of former students in the top 1% does not detectably exceed its association with median earnings (SI Section S8).

Table 2: Coupling under institution-level controls. Panels A and B: US, College Scorecard median earnings four years after completion unless stated, published field prestige F . Panel C: UK, LEO median earnings five years after graduation (cohorts 2013/14–2016/17), academia-wide UK hiring prestige G . k = fields (subjects). Intervals (95%): joint institution bootstrap, conditional on the observed fields (A); institution cluster bootstrap (B); two-stage bootstrap over subjects and providers (C); † : two-way (field, institution) or (subject, provider) cluster variance.

Specification	k / sample	Estimate [95% interval]
<i>A. US: mean within-field coupling of F with earnings</i>		
Raw, canonical sample	52	+0.419 [+0.355, +0.482]
Ladder (point estimates, common sample): raw $\rightarrow$ +	44	+0.43 $\rightarrow$ +0.31 $\rightarrow$ +0.24 $\rightarrow$
Pell share, control, state level $\rightarrow$ + admission rate $\rightarrow$ +		+0.14 $\rightarrow$ +0.12
SAT $\rightarrow$ + institution-wide earnings		
Broad set (SAT, admission rate, Pell share, control, state level)	46	+0.141 [+0.075, +0.208]
broad set dated to entry years (falls 2014–15) / 2026	45; 2,787	+0.134 [+0.065, +0.203] /
file, same programs †	programs	+0.129 [+0.062, +0.197]
Broad set + G (field-specific remainder)	44	+0.059 [+0.003, +0.114]
Broad set + G + institution-wide earnings	44	+0.047 [−0.011, +0.106]
Broad set + G , mean of 4-, 1- and 5-year earnings	39	+0.052 [+0.003, +0.100]
<i>B. US: within-institution slope, change in log earnings per within-field SD of F</i>		
Field fixed effects only	3,416 programs	+0.084 (SE 0.006)
+ institution fixed effects	3,416 programs	+0.0129 [+0.0048, +0.0212]
+ field-specific slopes on SAT, admission rate, Pell share, G (main)	2,889 programs	+0.0084 [+0.0000, +0.0164]
same programs, field fixed effects only	2,889 programs	+0.0851 (point)
same programs, field-specific slopes on SAT and G	2,889 programs	+0.0118 [+0.0037, +0.0205]
only		
main + field-specific local demand (ACS field $\times$ state earnings)	2,889 programs	+0.0084 [+0.0002, +0.0165]
main, selectivity dated to entry years / 2026 file †	2,801 programs	+0.0081 [+0.0005, +0.0158] /
		+0.0090 [−0.0008, +0.0187]
main, 1-year earnings (4-year on the same programs +0.0123)	2,681 programs	−0.0024 [−0.0150, +0.0101]
main, 5-year earnings (4-year on the same programs +0.0111)	2,549 programs	+0.0023 [−0.0069, +0.0119]
main, mean of 4-, 1-, 5-year earnings: unweighted / weighted by graduates	2,422 programs	+0.0048 [−0.0024, +0.0120] /
		+0.0108 [+0.0035, +0.0181]
main + field-specific program Pell share, composition sample	1,294 programs	+0.0137 [+0.0027, +0.0238]
same, Pell-recipient / other aided graduates' median as outcome	1,294 programs	+0.0102 [−0.0024, +0.0222] /
		+0.0009 [−0.0123, +0.0128]
<i>C. UK: coupling of G with earnings and prior attainment</i>		
Raw	33	+0.486 [+0.398, +0.559]
Raw, composition sample	32	+0.497 [+0.406, +0.571]
standardised for band mix, within-provider gradients (85% [80, 90] survives)	32	+0.425 [+0.335, +0.497]
standardised for band mix, national band medians (floor; 62% [56, 69])	32	+0.310 [+0.234, +0.374]
log-earnings gradient per SD of G , raw $\rightarrow$	32	+0.100 $\rightarrow$ +0.073 [+0.051,
within-provider standardised (73% [67, 78]) †		+0.095]
partial on institution intake selectivity (26% [17, 38]; two-way [+0.034, +0.229])	32	+0.131 [+0.078, +0.188]
Within prior-attainment band	28	+0.227 [+0.160, +0.279]
all graduates, same providers (66% [51, 79] survives; noise-corrected on 27 subjects 87% [74, 100], against 70% uncorrected there)	28	+0.342 [+0.255, +0.398]
within band: G net of intake selectivity †	28	+0.033 [−0.047, +0.112]
within band: intake selectivity net of G	28	+0.252 [+0.172, +0.310]

4.3 UK: most coupling survives a direct prior-attainment control

The UK data add what the US data lack, earnings by graduates' own prior attainment, but they have no department axis, so they bear on institution-level status against student attainment, not on the department question.

UK coupling of G with LEO earnings five years after graduation averages $+0.486$ [$+0.398$, $+0.559$] across 33 subjects, similar to the US coupling of G with Scorecard earnings ($+0.450$, 57 fields). Most of it survives a control for attainment (Table 2, panel C; Fig. 3). Among same-band graduates across providers coupling is $+0.227$ against $+0.342$ for all graduates of the same providers (28 subjects), so 66% [51, 79] survives, and 87% [74, 100] once the lower reliability of band medians is corrected for (27 subjects). Standardising each provider's earnings for its band mix, with band gradients measured within providers, leaves 85% [80, 90] (32 subjects). These shares describe orderings; in pay terms less survives, 73% [67, 78] of the log-earnings gradient in G . Coarse bands, sorting within a band, and within-provider gradients identified from students admitted despite a lower band all push the shares up (SI Section S5).

Among graduates of the same attainment band, what survives tracks intake selectivity at least as closely as hiring-network prestige if G is measured with little error: G adds $+0.033$ beyond selectivity (two-way [-0.047 , $+0.112$]), and selectivity keeps $+0.252$ [$+0.172$, $+0.310$] beyond G . On all-graduate medians G keeps more: net of selectivity $+0.131$ [$+0.078$, $+0.188$], 26% of the raw coupling, a fall comparable to the US one. This is the pattern Britton et al. (2022) report with a composite league-table rating, found again with a hiring-network axis. The UK G may measure standing with error: at a validity of 0.74 the data allow a same-band partial up to $+0.26$ (SI Section S5). Subjects differ, and the differences survive the attainment control, but subjects whose pay couples strongly with G are those whose pay couples strongly with intake selectivity (Spearman $+0.92$ [$+0.71$, $+0.95$]); as in the US, the subject ordering cannot be separated from subject-specific selectivity gradients (SI Section S5).

4.4 Coupling rises with years since graduation

Within fixed graduation cohorts, coupling rises with years since graduation in both countries. In the US it also rises across graduation cohorts, much of the within-cohort rise is shared with selectivity, and in a weakly identified decomposition it appears in the loading on academia-wide prestige G , not on field prestige F .

US within cohorts. On the balanced PSEO panel, mean coupling rises from $+0.173$ one year after graduation to $+0.310$ at five years and $+0.451$ at ten (39 fields, V4.13.0; Fig. 4a). The mean per-field slope is $+0.031$ per year (two-way interval [$+0.021$, $+0.041$]; $+0.030$ on V4.14.1, Table 3). It is positive in each of the four cohorts and in both segments, so it does not rest on the first-year point, and correcting the program medians for sampling noise does not shrink it (exploratory; SI Section S6).

Cohorts and calendar time. A within-cohort change mixes career time with calendar time. On the 17-cell surface of graduation windows by horizon (24 fields whose institutions are present in every cell; V4.14.1), coupling rises both within windows ($\theta_h = +0.031$ [$+0.019$, $+0.043$] per year since graduation) and across windows at a fixed horizon ($\theta_c = +0.015$ [$+0.008$, $+0.023$]; Fig. 4b). Linear career (a), calendar (b) and cohort (g) trends are not identified, since $\theta_h = a + b$ and $\theta_c = b + g$; if calendar and cohort trends are both non-negative, the career-time slope

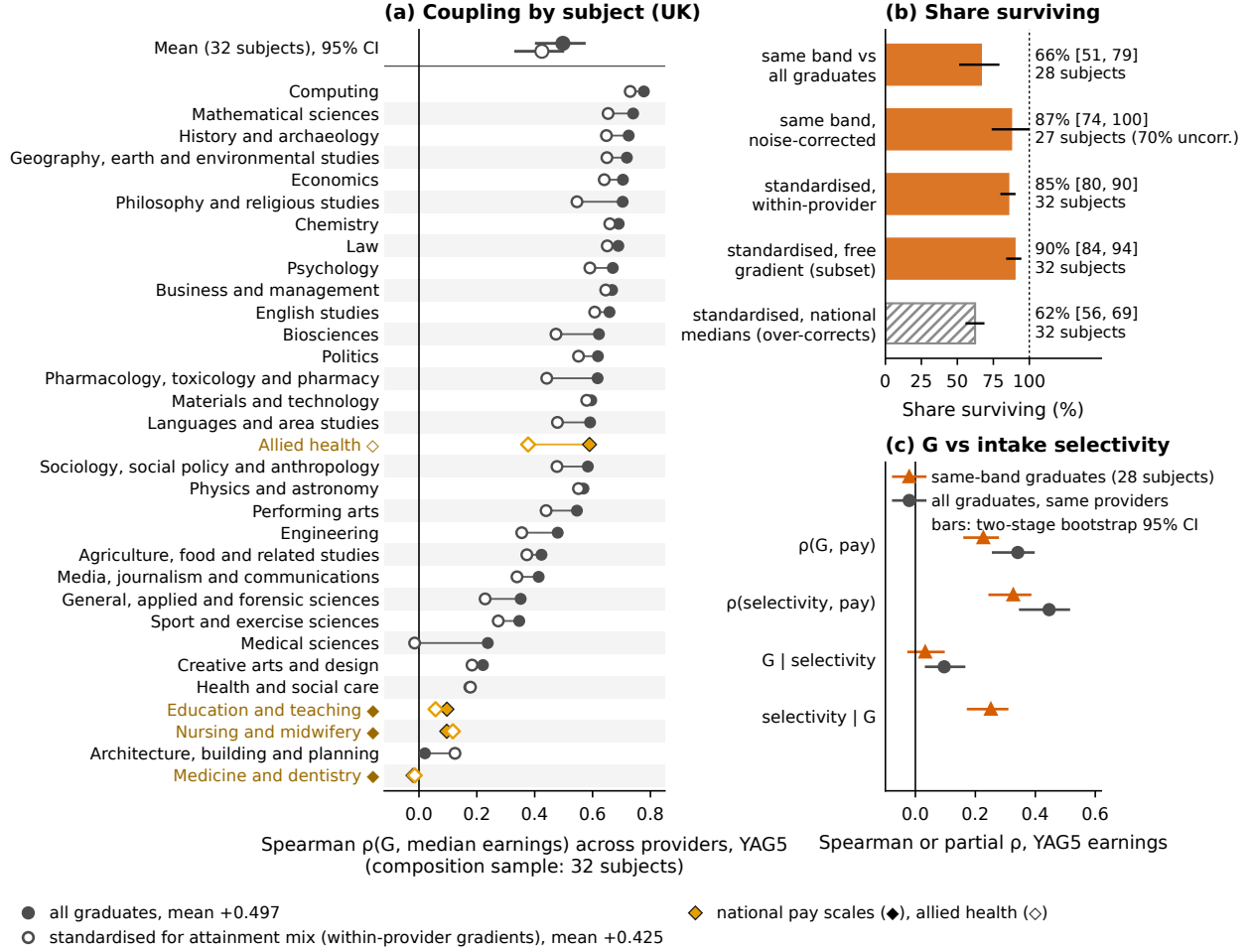


Figure 3: UK: coupling and prior attainment (LEO five-year earnings, cohorts 2013/14–2016/17). (a) Coupling of G by subject, all graduates (filled) and standardised for attainment mix (open), on the 32-subject composition sample, whose values differ from those on all providers (Fig. 5b). (b) Share of coupling surviving each attainment control. (c) Coupling of G and of intake selectivity with earnings, and each net of the other, for same-band (triangles) and all graduates (circles). Two-stage bootstrap 95% intervals; two-way intervals of $G | \text{selectivity}$: $[-0.047, +0.112]$ and $[-0.003, +0.195]$.

lies in $[\theta_h - \theta_c, \theta_h] = [+0.015, +0.031]$ per year (Fosse and Winship, 2019b), and the calendar-matched contrast on the balanced panel gives nearly the same bound (Table 3). A zero career-time slope would need a calendar trend of about +0.031 and a cohort trend of about −0.015 per year. The bound also assumes that graduate study does not depress the early medians of higher-status institutions more than others'; under that confound both ends are biased upward. Prestige is measured once, from 2011–2020 hires, so any drift as cohorts approach that period enters b or g . Two tests of entry conditions fixed before estimation, aggregate analogues of Oreopoulos et al. (2012), are not informative (SI Section S6).

Selection into the data, graduate study and location. Higher-prestige institutions have a smaller share of graduates with PSEO earnings at y_1 ; partialling coverage out removes about a fifth

of the slope (Table 3; cf. Foote and Stange, 2026). Graduate study could raise coupling mechanically if graduates of higher-status institutions more often enter graduate programmes, drop out of the earnings sample while studying and return with higher pay. With coverage and the institution’s research-doctorate rate per bachelor’s graduate as controls, the rise is $+0.0103$ [$+0.0027$, $+0.0178$] per year, against $+0.0305$ raw on the same 144 cells; but the doctorate rate is institution-wide and tracks G (within-cell Spearman $+0.71$), so the removed share mixes timing with a status gradient. At the field level the mechanism is neither shown nor excluded (power 0.31). About one fifth to one third of the rise moves with the wage level of the census divisions where graduates work and with the out-of-state share (post hoc, self-attested registration). The rise survives every available proxy; its size does not (SI Section S6).

Selectivity and status loadings. Net of SAT and admission rate dated to the cohorts’ entry years, the within-cohort slope is $+0.011$ [$+0.003$, $+0.020$] per year, 0.39 of the raw $+0.029$ on the same 109 cells (V4.14.1): about three fifths of the rise is shared with selectivity. The loading of pay on G net of selectivity (SAT, admission rate, Pell share, control and state, and F in the rank regression) does not decline within cohorts: its slope is $+0.016$ [-0.011 , $+0.044$] per year in the rank regression, which excludes a decline steeper than 30% of the uncontrolled rise, and $+0.014$ [$+0.003$, $+0.024$] as a partial correlation (37 fields, V4.14.1). This test takes selectivity from the 2026 file, whose admission rates rank the PSEO institutions only loosely like those of their entry years (Spearman 0.45–0.47; SAT 0.88–0.90). Mismeasured selectivity leaves part of itself in G , whose loading can then rise, so the dating favours finding no decline. Entry-year SAT averages and admission rates exist for these cohorts (Pell shares do not), but the loading test was not re-run with them. Selectivity is also public and learnable from earlier cohorts, so individual employer learning need not produce an aggregate decline. In the UK, G ’s loading net of selectivity is flat ($+0.008$ [-0.006 , $+0.022$], two-way) while the selectivity loading rises ($+0.024$ [$+0.011$, $+0.037$]); their difference includes zero, and with a negative UK cross-cohort drift these are not career-time statements. Neither country shows the decline that employer learning predicts once the correlate is held fixed (Altonji and Pierret, 2001), consistent with little post-entry learning about college graduates (Arcidiacono et al., 2010) but not a test of it.

Academia-wide against field prestige. In a standardised rank regression of earnings on F and G within each field-by-cohort cell, b_F stays flat ($+0.189$ at y_1 , $+0.177$ at y_{10}) while b_G grows from -0.013 to $+0.328$ (Fig. 4d). Their slope difference $d_G - d_F$ is $+0.039$ [$+0.009$, $+0.069$] under the two-way variance (V4.13.0), but the conservative crossed bootstrap reaches zero, and the decomposition is weakly identified: within cells F and G correlate 0.83, and G is measured more reliably. In a pooled form of the regression, and net of selectivity, G ’s lead over F is not distinguishable from zero (SI Section S6). What the growing G -tracked component is (residual selectivity, reputation, resources, location or the timing of graduate study) is not identified.

UK. On fixed LEO cohorts (33 subjects), coupling rises from $+0.379$ at YAG1 to $+0.484$ at YAG5, $+0.026$ per year [$+0.014$, $+0.037$] (Fig. 4c), in all four cohorts and both segments, and by $+0.028$ [$+0.016$, $+0.038$] after within-provider standardisation for prior attainment. The calendar-matched contrast ($+0.031$ [$+0.021$, $+0.042$]) exceeds the within-cohort change on the same providers ($+0.025$), so the sign assumption of the US bound holds only within sampling error, and offsetting calendar and cohort effects cannot be excluded (SI Section S5).

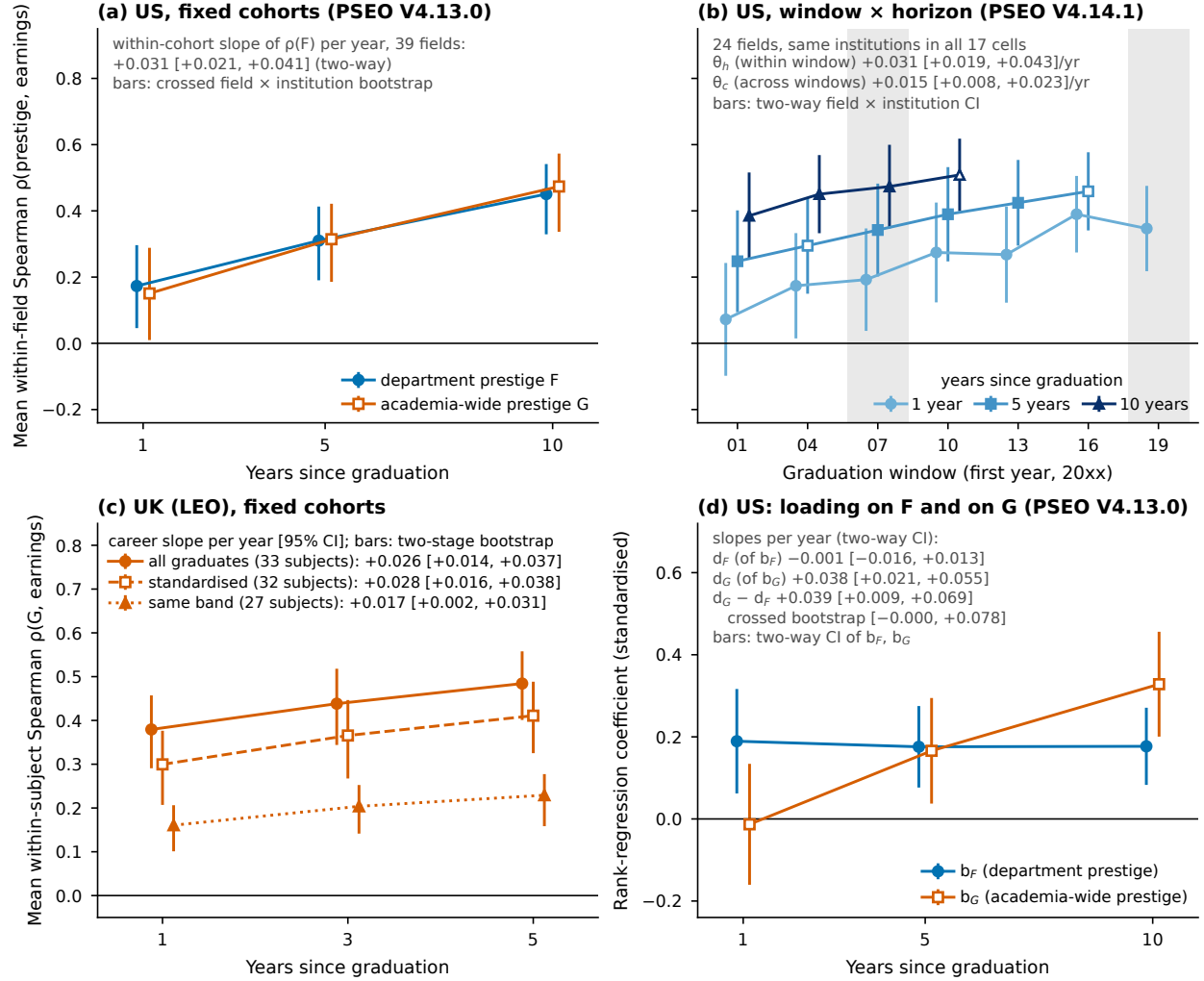


Figure 4: Coupling and years since graduation. (a) US: mean coupling of F and G with PSEO medians at one, five and ten years, fixed cohorts 2001–03 to 2010–12 (39 fields, V4.13.0). (b) US: coupling of F in 17 graduation-window $\times$ horizon cells (24 fields, V4.14.1); shading: earnings in 2008–10 or 2020–22. (c) UK: coupling of G , fixed LEO cohorts 2013/14–2016/17: all graduates, standardised for attainment mix, within bands. (d) US: rank-regression loadings on F and G , panel of (a); weakly identified. Within-cohort changes mix career and calendar time. 95% intervals, types stated in each panel.

Reading. The direction agrees with descriptive evidence (Thomas and Zhang, 2005; MacLeod et al., 2017). Designs that compare students admitted to the same colleges find a selectivity premium that is small over the career or fades within a few years (Dale and Krueger, 2014; Mountjoy and Hickman, 2021); a descriptive gradient can rise while such a premium fades. Given that and the three fifths shared with selectivity, the most parsimonious reading is residual selection: growing pay returns to pre-college differences that status and selectivity proxy. Holding both fixed, institutions with more Pell students lose their early relative position (SI Section S6), which growing returns to family advantage, or graduate study, would produce. A rising association also fits a growing return to status, sorting into careers with steeper earnings profiles (Binder et al., 2016; Rivera, 2015), and the timing of graduate study. Aggregate data cannot separate these readings.

Table 3: Career-time summary. US: PSEO release V4.14.1, on which every row is available (the canonical V4.13.0 gives slopes within 0.004 per year of these; SI Section S6); bachelor’s graduates, field prestige F , per-field slopes averaged over fields, two-way (field, institution) cluster intervals unless marked; balanced fixed cohorts 2001–03 to 2010–12 unless the row states another sample. UK: LEO, balanced fixed cohorts 2013/14–2016/17, academia-wide prestige G ; two-stage bootstrap over subjects and providers. Slopes are changes in coupling (or in a loading) per year since graduation. 95% intervals.

Quantity	US, V4.14.1 (40 fields) ^a	UK (33 subjects)
Coupling at first / middle / last horizon	+0.159 / +0.294 / +0.433	+0.379 / +0.438 / +0.484
Within-cohort slope (crossed bootstrap)	+0.030 [+0.020, +0.041] ([+0.018, +0.043])	+0.026 [+0.014, +0.037]
Calendar-matched contrast; US without earnings years 2008–10, 2020–22 (38 fields)	+0.015 [+0.007, +0.023]; +0.014 [+0.005, +0.024]	+0.031 [+0.021, +0.042]
Cross-cohort drift at entry	+0.018 [+0.010, +0.025]	−0.006 [−0.015, +0.003]
Career-time bound if calendar, cohort effects ≥ 0 ^b	[+0.015, +0.032]	—
17-cell surface (24 fields): θ_h / θ_c	+0.031 [+0.019, +0.043] / +0.015 [+0.008, +0.023]	—
Coverage control: share of slope removed (US); slope net of coverage (UK)	0.195 [0.097, 0.293]	+0.017 [+0.005, +0.027]
Slope net of coverage and research-doctorate rate (144 cells; raw +0.0305)	+0.0103 [+0.0027, +0.0178]	—
Slope net of entry-year SAT and admission rate (39 fields, 109 cells; raw +0.0289)	+0.0112 [+0.0026, +0.0198]	—
Prior-attainment standardised slope; with G corrected for validity 0.74	—	+0.028 [+0.016, +0.038]; +0.035 [+0.019, +0.051] ^c
Slope of b_G (d_G) / of b_F (d_F)	+0.035 [+0.019, +0.052] / +0.000 [−0.013, +0.014]	—
$d_G - d_F$ (crossed bootstrap)	+0.035 [+0.007, +0.063] ([−0.003, +0.070])	—
Loading on G net of selectivity (2026 controls): rank regression / partial correlation (37 fields)	+0.016 [−0.011, +0.044] / +0.014 [+0.003, +0.024]	+0.008 [−0.006, +0.022] ^c

^a The calendar-matched contrast, the drift and the bound use the 39 fields with triple-matched institutions; rows naming another sample use it. ^b Both ends would be biased upward if graduate study depressed the early medians of higher-status institutions more than others’. ^c Two-way (subject, provider) interval.

4.5 Where coupling is weak: national pay scales and strictly licensed occupations

UK. Nursing and midwifery, medicine and dentistry, and education and teaching, three subjects whose graduates are largely paid on national scales, couple at +0.120 [−0.033, +0.258] against +0.523 [+0.439, +0.585] for the other 30 subjects (difference −0.404 [−0.566, −0.238]; exact subject-label permutation $p = 0.0009$; Fig. 5b), and the contrast survives standardisation for prior attainment (SI Section S5). Allied health, much of it also paid on national bands, is an exception at +0.570. The pattern and its reading as centrally regulated pay are Britton et al.’s (2022); we add nursing and a hiring-network axis.

US. Across US fields the prestige–pay association net of geography falls with the share of a field’s graduates in strictly licensed occupations (health practitioners, school teachers, lawyers, architects and psychologists; ACS 2023): Spearman −0.53 to −0.48 over two institution samples and three geography codings (Fig. 5a). It is a contrast between discipline families (engineering, computer science, statistics and business at the low-share end), and it is fragile: under resampling of whole CIP-2 families its interval excludes zero in only 3 of 6 versions, it is not detected within families or with a broad licensure share, a government-or-non-profit employment share predicts it at least

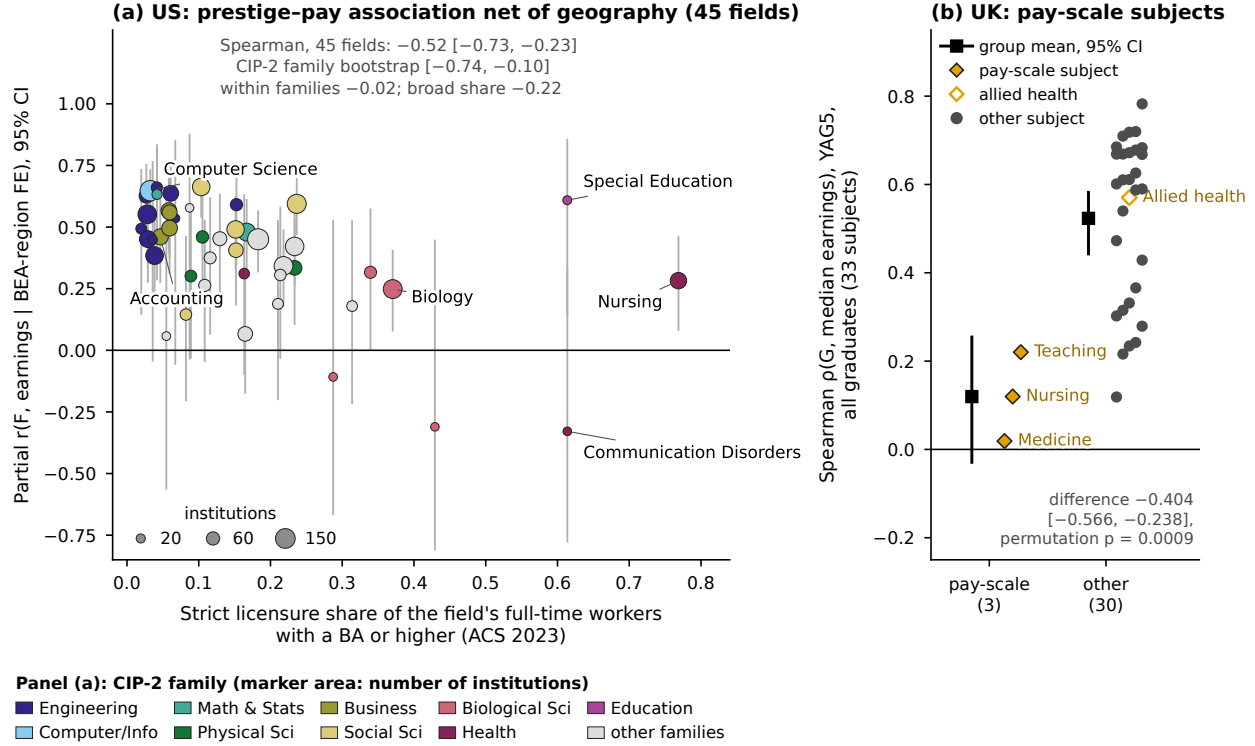


Figure 5: Where coupling is weak. (a) US: partial correlation of F with Scorecard four-year earnings net of BEA-region effects (one of six sample-by-geography versions) against the field's share of full-time bachelor's holders in strictly licensed occupations (ACS 2023); one point per field, coloured by CIP-2 family, sized by institutions. (b) UK: coupling of G with LEO five-year earnings (33 subjects) in the three pay-scale subjects, allied health and the others; group means. 95% intervals.

as well, and special education, whose graduates mostly teach on district salary schedules, couples strongly. Pay set by the setting, such as a teacher salary schedule (Biasi, 2021) or a pay scale, is a reading consistent with these contrasts; whether licensure itself raises wages is contested (Kleiner and Krueger, 2013; Redbird, 2017), and these data isolate neither reading (SI Section S7).

5 Discussion

For the science of science. The faculty-hiring hierarchy is a reliable measure of standing inside the academy, the academia-wide ordering especially, and it does track the pay of the academy's bachelor's graduates. Outside academia, however, the information it carries about their pay is mostly institution-level status. Within institutions one SD of department prestige goes with 0.8% higher four-year pay, with an interval that reaches zero (about zero with one- or five-year earnings; up to about 3% corrected for measurement error at the lower reliability bound); a field's own ordering predicts pay no better than the academia-wide ordering; and in the UK, among graduates of the same attainment band, the academia-wide ordering adds no detectable amount beyond intake selectivity. Because G pools the same hiring data over all fields, "institution-level status" here is the part of the hierarchy shared across an institution's departments. Studies that use hiring-network prestige as a proxy for a program's training quality or labour-market standing should expect it to behave mostly as a measure of institutional status. The one field-specific exception, computer

science, fits selection into the major as well as department standing, and it is measured without the University of California campuses and several flagship public departments.

For the college-quality literature. A hiring-network axis shows the pattern Britton et al. (2022) report: most of the prestige ordering of pay goes with selectivity (US +0.43 to +0.14 across about 50 fields; UK +0.50 to +0.13 on 32 subjects), and what remains beyond selectivity and academia-wide status is small. Unlike their league-table result, 11 of 46 US fields keep a partial coupling more than 1.96 standard errors above zero net of selectivity alone. The cross-field ordering of coupling is nearly collinear with how steeply each field’s pay follows institutional selectivity, a gradient that itself varies across fields (Bloem et al., 2024). At the resolution of public data, “the status gradient in pay differs by field” and “the department-prestige gradient in pay differs by field” cannot be told apart; the within-institution design narrows the gap between them to a small remainder.

Career time. Within fixed cohorts, coupling rises by about +0.03 per year in two countries with different earnings sources, prestige measures and horizons. In the US it also rises across cohorts, so the career-time part is bounded, [+0.015, +0.031] per year, only under a sign assumption and if graduate study does not depress the early medians of higher-status institutions more than others’; about three fifths of the within-cohort rise is shared with selectivity; and what remains appears, in a weakly identified decomposition, in the loading on the academia-wide ordering. Net of selectivity the status loading does not decline within cohorts in either country; in the US selectivity is measured on recent entrants, a dating that favours this result. Set against small or fading causal selectivity premia (Dale and Krueger, 2014; Mountjoy and Hickman, 2021), the most parsimonious reading is residual selection, growing returns to pre-college differences that status and selectivity proxy; the pattern also fits employers who know at hiring most of what status and selectivity signal, sorting into steeper careers, a growing return to status and the timing of graduate study. The data do not distinguish these.

What would decide the open questions. Four kinds of data would separate department standing from program-level selection: program-level entry attainment, which no public US file has; changes in department standing over time, pairing department ratings from different assessment rounds (National Research Council, 1995, 2011) with PSEO cohorts; admission routes into majors for a larger sample, including engineering, business and the top departments now lost to the name join or the missing SAT average; and linked student records with entry scores, major and earnings, the decisive design, which are not public. Two re-estimates are feasible with the public files already used: the within-institution and computer-science analyses with a join by institution identifier and entry-year SAT averages, and the status-loading test with entry-year SAT averages and admission rates. For career time, graduate enrolment by institution, field and cohort would test the graduate-study reading directly.

6 Limitations

Each of the following limits a specific claim. *Medians and coverage.* Scorecard medians describe Title IV-aided completers who work and are not enrolled; PSEO lacks most of the top prestige quartile, and neither source sees the upper tail within a program. *Sample.* The name join and the missing fall-2024 SAT averages of test-blind institutions remove 18 of the 50 highest-ranked

computer-science departments from the four-year within-institution analyses (21 from the pooled-horizon computer-science sample), among them the University of California campuses, Caltech, Georgia Tech, Purdue, Ohio State and the University of Washington; these analyses were not re-run with a join by institution identifier or with entry-year SAT averages. *Selectivity proxies.* Selectivity is institution-level; re-dating it to entry years moves none of the three primary statistics detectably (only the Pell-share step of the ladder), but cannot show that the proxies capture student quality, and whatever they miss stays in every partial coupling. The status loadings, the +0.059 remainder and the +0.94 use 2026 controls; for the status loadings that dating favours the reported absence of a decline. *Location.* The local-demand control describes the institution’s state, passes little through to program earnings and mixes demand with who lives there. *Earnings reliability.* The noise propagation is secondary, not yet independently verified, and its error model is approximate. *Timing.* Career dynamics are not separable from calendar time without a sign assumption, or from the timing and returns of graduate degrees, which could bias both ends of the bound upward. *Prestige.* The reliability of the published field ranks can only be bracketed (0.81–0.96), so the size of the remainder is uncertain (+0.0098 to +0.0297 disattenuated); *F* partly tracks doctoral production; and the UK hierarchy rests on self-reported ORCID careers of uncertain validity. *Rank statistics.* Coupling and the UK survival shares describe orderings; low coupling in UK pay-scale subjects may reflect little cross-provider pay variation. *Descriptive design.* Nothing here separates value added, selection and employer screening, and the employer-learning analogue uses institution aggregates and a public correlate. *Field-level inference.* Moderators measured once per field rest on 40–50 fields and about 18 CIP-2 families. *Power.* Nulls are statements of what the data cannot detect; SI Section S9 gives the power or interval of the main nulls. *Beliefs* about prestige are not measured. *Specification.* The canonical specification was fixed after the main analyses had been run; SI Section S4 shows how the main statistics move with the alternatives.

7 Methods

Within-institution model. For the program of field f at institution i ,

$$\log Y_{if} = \alpha_i + \gamma_f + \beta z(F)_{if} + \sum_{f'} D_{f'} (d_{f'} z\text{SAT}_i + a_{f'} z\text{ADM}_i + p_{f'} z\text{Pell}_i + t_{f'} zG_i) + e_{if}, \quad (1)$$

where $z(\cdot)$ standardises within field, α_i and γ_f are institution and field fixed effects, and $D_{f'}$ indicates field f' . The field-specific slopes let the earnings gradients in selectivity, Pell share and G differ by field, so β is identified only from departments whose standing differs from what their institution’s traits imply. Institutions with one field and fields with fewer than 15 programs are dropped iteratively; intervals come from an institution cluster bootstrap with 999 draws shared across specifications, so that differences are paired (SI Section S4).

Career-time contrasts. On institutions observed in all three cells, the within-cohort change $(y_{10} - y_1)/9$, the calendar-matched contrast (cohort c at y_{10} against cohort $c + 9$ at y_1 , over 9) and the cross-cohort drift at y_1 equal $a + b$, $a - g$ and $b + g$ in a linear model with career, calendar and cohort effects a , b and g ; on the 17-cell surface, the within-window and across-window slopes equal $\theta_h = a + b$ and $\theta_c = b + g$. Only under $b, g \geq 0$ is the career-time effect bounded: $a \in [a - g, a + b]$, that is, between the calendar-matched contrast and the within-cohort change, or $a \in [\theta_h - \theta_c, \theta_h]$ on the surface (Fosse and Winship, 2019b). No point estimate of a is reported (SI Section S6).

Further details. The SI documents data and crosswalks (SI Section S1), prestige measures (SI Section S2), the coupling estimator and earnings reliability (SI Section S3), the selectivity designs (SI Section S4), the UK analysis (SI Section S5), career time (SI Section S6), licensing and employer sector (SI Section S7), tail outcomes (SI Section S8), power and the hypotheses examined (SI Section S9), and earnings as a placement measure (SI Section S10).

Declarations

Availability of data and materials. All inputs are public: the faculty-hiring data (Wapman et al., 2022a); College Scorecard field-of-study, institution and historical institution files (U.S. Department of Education, 2026); Census PSEO earnings and flows, releases V4.13.0 (2025Q4) and V4.14.1 (2026Q2) (U.S. Census Bureau, Longitudinal Employer-Household Dynamics, 2025, 2026); the Department for Education’s “Graduate outcomes (LEO): provider level data”, 2022-23 release (Department for Education, 2026); the ORCID-derived academic mobility edges (version 20260419, doi:10.5281/zenodo.19651302; Li, 2026), built from ORCID public data (ORCID, Inc., 2026), with institutions resolved through the ROR data dump (v2.8, doi:10.5281/zenodo.20512981; Research Organization Registry, 2026); Opportunity Insights college tables, including the Barron’s 2009 selectivity index (Chetty et al., 2020); the 2023 American Community Survey public-use microdata (U.S. Census Bureau, 2026); the Survey of Earned Doctorates by baccalaureate institution (National Center for Science and Engineering Statistics, 2026) and IPEDS completions and directory files (National Center for Education Statistics, 2026); state unemployment rates from the Bureau of Labor Statistics Local Area Unemployment Statistics (U.S. Bureau of Labor Statistics, 2026); occupational prestige ratings (Hughes et al., 2024; Condon and Hughes, 2022) and the New York Fed college labour-market data by major (Federal Reserve Bank of New York, 2026); and official university admissions pages, saved with their access dates. Provenance for every input (URL, version, access date and licence) is recorded in `data/raw/SOURCES.md` of the repository. No input dataset is redistributed; the derived tables are regenerated by the scripts. The hand-coded computer-science admission file (URL, access date and verbatim sentence for each program), which no script can regenerate because the pages change, and the small derived tables (UK prestige ranks, CIP map) are available from the author.

Code availability. The numbers in this article come from seeded scripts (scripts 01–75; mainly 52–75). The figures, and the numbers marked “SI check”, come from `scripts/70_paper_figures.py` (with `-si-checks`); the SI names the source of every other number. The repository is at <https://github.com/soray42/degrees-of-separation>. At the time of writing, scripts 67–69 and 72–75 and the current versions of scripts 65 and 70 are not yet in the public repository; they will be added, with the digests of their pre-specification blocks, in a tagged release.

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